Assignment No 1

Title : To implement a distributed data processing pipeline using Apache Hadoop MapReduce and Apache Spark and compare their performance.

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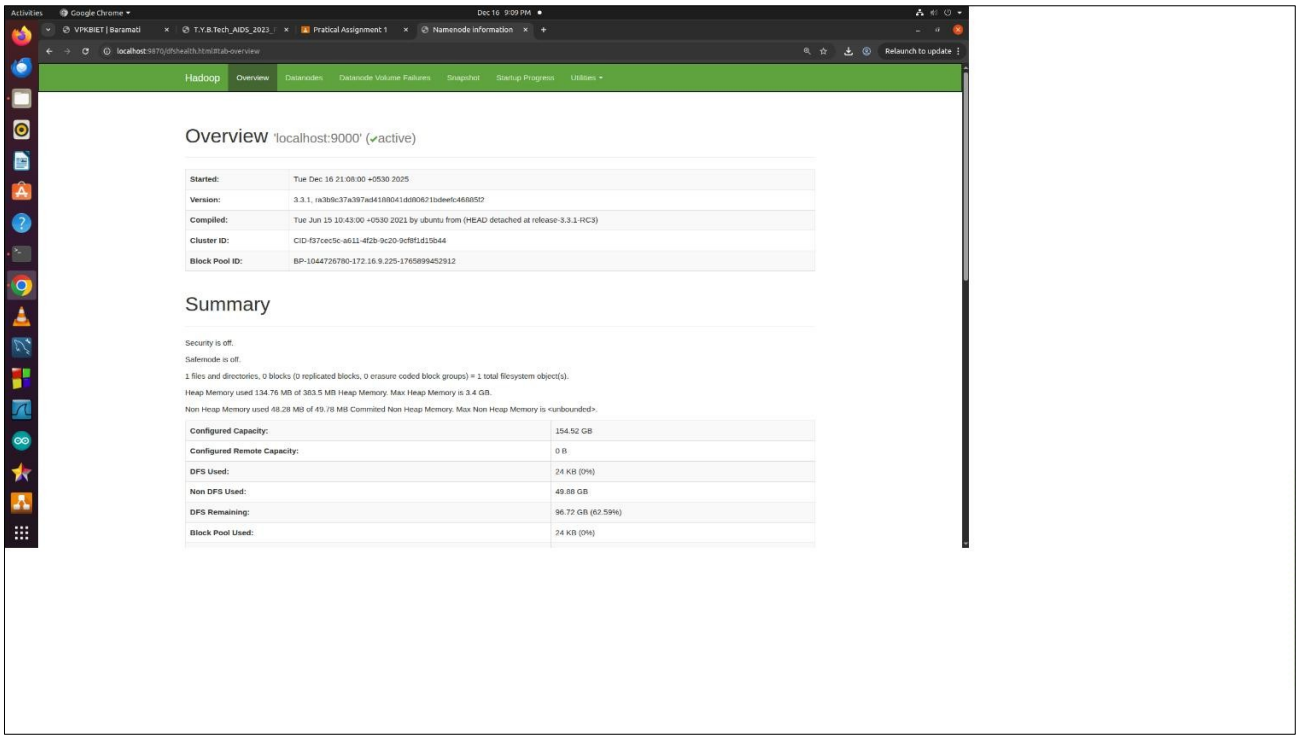
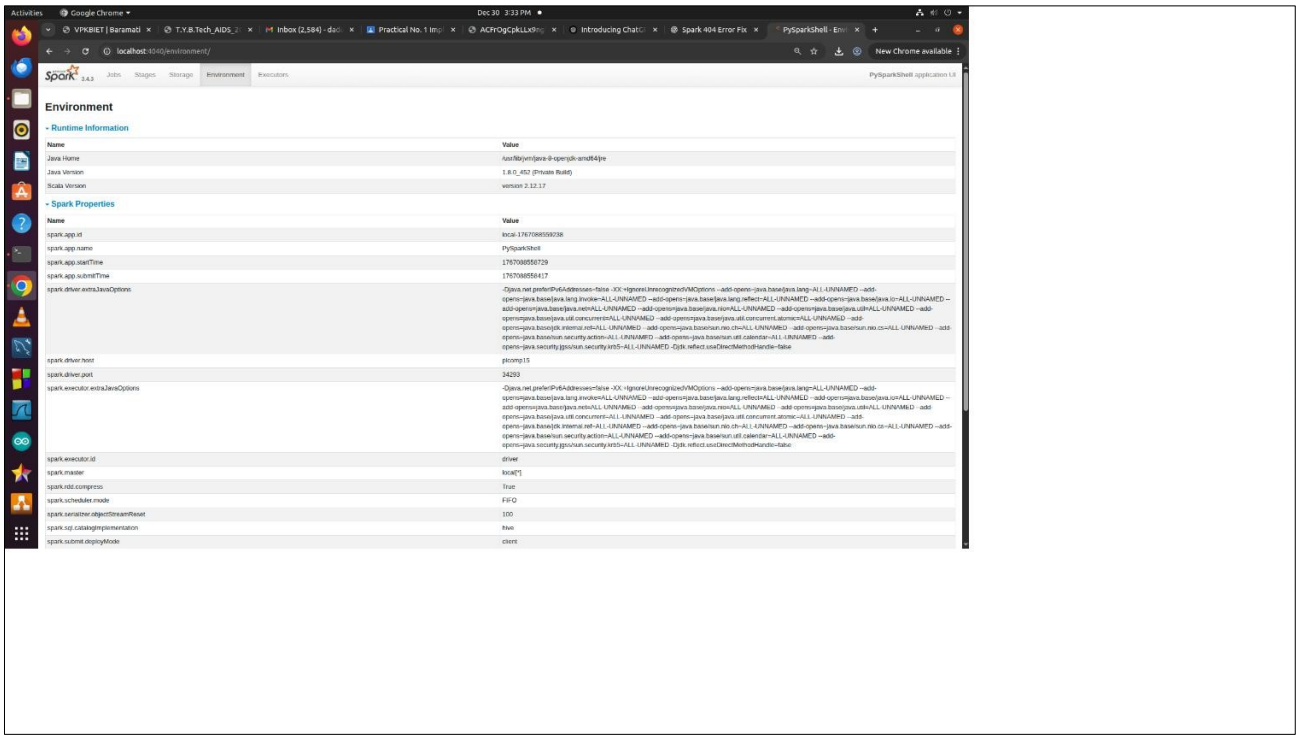
1) Hadoop :

sudo apt-get update
ssh localhost
hadoop-3.3.1/bin/hdfs namenode -format
start-all.sh
Open the Browser and Type:- localhost:9870



2) Spark :
python3 --version
which python3
nano ~/.bashrc
export PYSPARK_PYTHON=/usr/bin/python3
export PYSPARK_DRIVER_PYTHON=/usr/bin/python3
source ~/.bashrc
cd /opt
sudo wget https://archive.apache.org/dist/spark/spark-3.5.0/spark-3.5.0-bin-hadoop3.tgz
sudo tar -xvzf spark-3.5.0-bin-hadoop3.tgz
sudo mv spark-3.5.0-bin-hadoop3 spark
nano ~/.bashrcpdf

export SPARK_HOME=/opt/spark
export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin
source ~/.bashrc
pyspark
Spark context Web UI available at http://localhost:4040